Supplementary Diagnostics for a Reproducible Binary-Schedule Low-Thrust Cislunar Benchmark

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Contents

Purpose

This supplement preserves secondary diagnostics, implementation-heavy tables, and small-seed evidence for the normalized Earth–Moon CR3BP missed-thrust benchmark. The main article reports the primary claim path: the claim hierarchy, the 30-seed duty-cycle-prior initializer comparison, the focused 30-seed QAOA/QUBO ablation statistics, and the strongest scoped hard-catalog tail-coast continuous-backend diagnostic. The evidence below is retained to make failure modes, provenance, caveats, and small-sample results auditable without making the main article a full artifact audit.

1 Catalog-Derived Schedule and QUBO Diagnostics

The original catalog-derived pilot remains negative evidence. QUBO simulated annealing and QAOA can produce competitive pre-refinement schedules in the small pilot, but no method meets the configured post-refinement robust thresholds after continuous recovery. The QUBO fit diagnostics show why surrogate validation is a first-order artifact rather than an assumption: with a full quadratic model and a small training set, training fit is much stronger than validation behavior.

Table 1: Catalog-derived pilot results. Values are medians across the generated pilot seeds. This benchmark did not meet the robust refinement thresholds.

Method	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
random	1.8309	2.5333	34.0442	0.3571	120.0000	0.0000	120.0000	0.0000
cross_entropy	0.9023	0.9031	9.9415	0.5000	120.0000	0.0000	120.0000	0.0000
genetic	2.5371	2.6132	34.8813	0.5000	132.0000	0.0000	132.0000	0.0000
true_sa	2.3540	2.4855	34.2025	0.6429	120.0000	0.0000	120.0000	0.0000
surrogate_qubo_sa	2.6497	2.7770	33.8472	0.5714	48.0000	72.0000	120.0000	0.0000
qaoa_statevector	2.4262	2.5573	33.8785	0.6429	48.0000	72.0000	120.0000	0.0000
all_windows_continuous	1.6383	2.2746	23.7862	1.0000	31.0000	0.0000	31.0000	0.0000

Table 2: Catalog-derived QUBO surrogate fit diagnostics.

seed	ridge	train_rmse	val_rmse	train_r2	val_r2	n_train	n_val	shared_qubo_training_evaluations	qubo_training_true_evaluations	equal_total_budget
11	0.0001	0.0002	4.3189	1.0000	0.5063	54	18	72	72	True
23	0.0001	0.0002	4.0060	1.0000	0.7190	54	18	72	72	True
37	0.0001	0.0002	5.9296	1.0000	0.0352	54	18	72	72	True

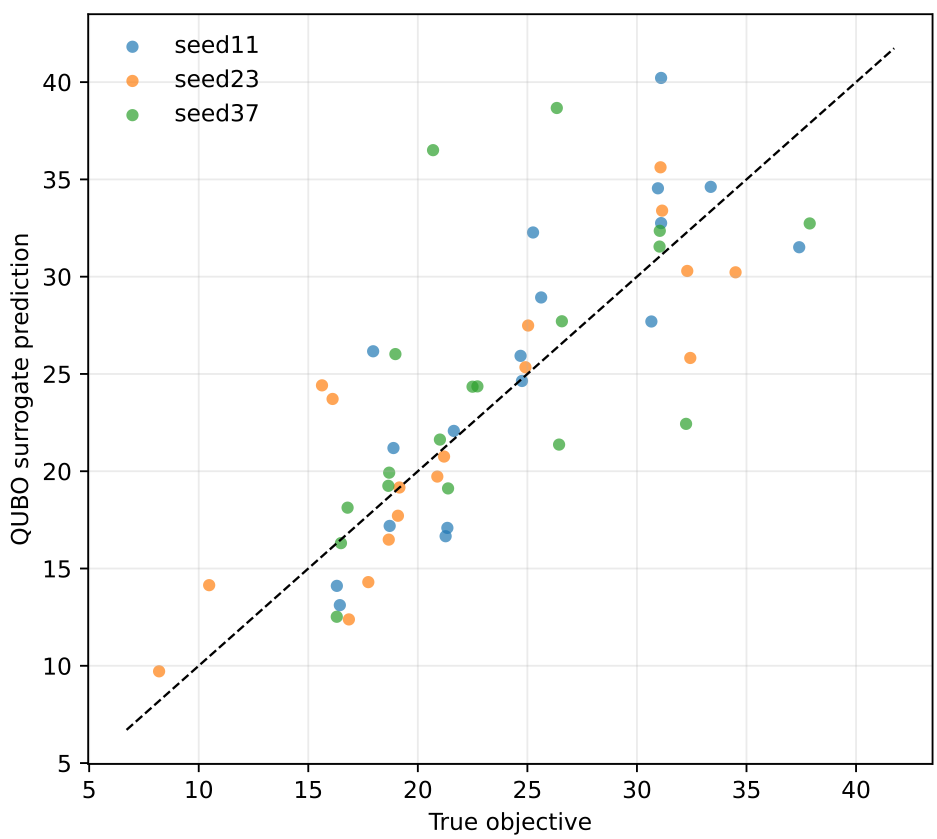

Figure 1: Catalog-derived QUBO fit diagnostic. The fit is reported as a diagnostic, not as evidence that the learned quadratic surrogate preserves the true trajectory ranking.

2 Catalog Feasibility and Multiple-Shooting Diagnostics

The feasibility sweep, direct multiple-shooting rows, targeted catalog attempt, and selected-outage envelope explain why the hard catalog-derived transfer is not promoted as a solved trajectory-design case. Selected recovery errors can be made small for chosen masks, but nominal errors, optimizer/backend success, and all-mask diagnostics remain limiting.

Table 3: Feasibility sweep stress test for the catalog-derived benchmark. None of the completed cases met both nominal and selected robust thresholds.

Transfer time	amax	Segments	Max nfev	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Meets thresholds	Best start	nfev	Runtime (s)
4.0000	0.2000	14	120	0.2008	0.2310	2.8975	False	feedback	251	1635.4279
4.0000	0.3000	14	150	0.1229	0.2783	7.5962	False	feedback	177	1102.4643
4.0000	0.3000	18	150	0.2423	0.2823	5.0379	False	feedback	39	351.0886
5.0000	0.3000	14	150	0.5810	0.6327	57.6216	False	zero	30	88.5080
5.0000	0.1500	14	120	0.6248	0.6785	53.6764	False	zero	138	997.2723
4.0000	0.2500	14	150	1.1558	1.2770	60.9139	False	feedback	81	324.7229
4.0000	0.1500	14	120	0.7389	1.6931	37.8087	False	zero	32	122.3204
3.0000	0.0650	14	120	1.0202	2.5174	7.1874	False	zero	61	326.9638

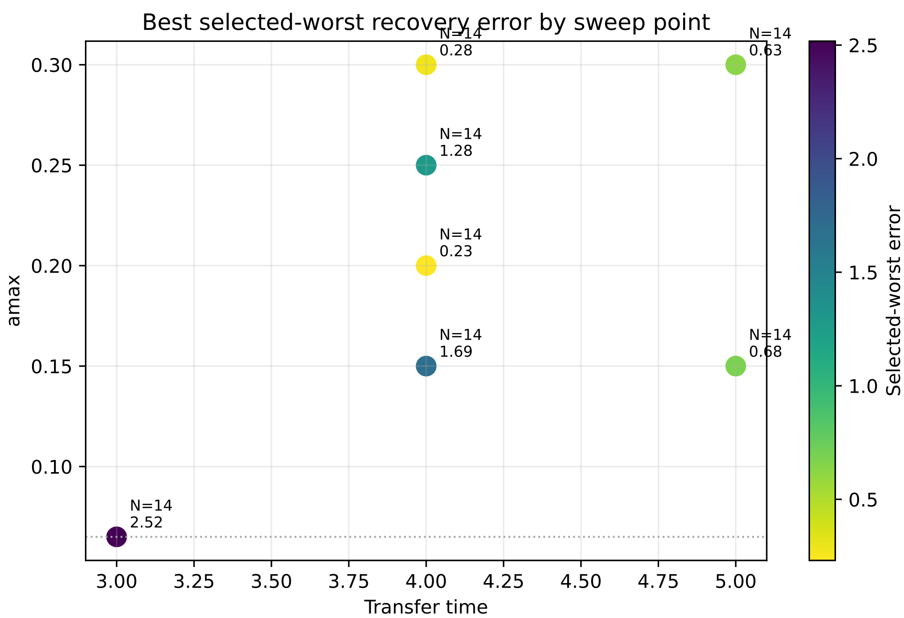

Figure 2: Feasibility sweep heatmap for the catalog-derived benchmark. The completed cases remain outside the configured robust success thresholds.

Table 4: Catalog-derived direct multiple-shooting diagnostics. Rows with zero selected outages are nominal-only diagnostics; all-outage diagnostic errors are still reported.

Transfer time	amax	Segments	Selected outages	Initialization	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Fuel	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds
4.0000	1.0000	14	1	blend	0.1714	0.0004	8.1180	4.0000	1.0000	0.0000	220	507.1263	False
5.0000	0.5000	14	0	linear	0.5154	0.5154	55.2845	2.5000	0.5000	0.0000	100	43.4353	False
4.0000	0.5000	14	3	linear	1.5355	1.2268	25.1426	2.0000	0.5000	0.0000	100	118.9605	False
4.0000	0.3000	14	3	linear	1.5981	1.9548	20.1304	1.2000	0.3000	0.0000	14	22.9601	False
4.0000	0.3000	14	3	linear	1.6273	4.3964	5.5503	1.2000	0.3000	0.0000	74	172.0316	False
5.0000	0.5000	14	3	linear	6.1753	5.7643	9.0650	2.5000	0.5000	0.0000	30	68.2684	False
6.0000	0.5000	14	0	linear	7.4830	7.4830	119.2412	3.0000	0.5000	0.0000	100	49.9072	False

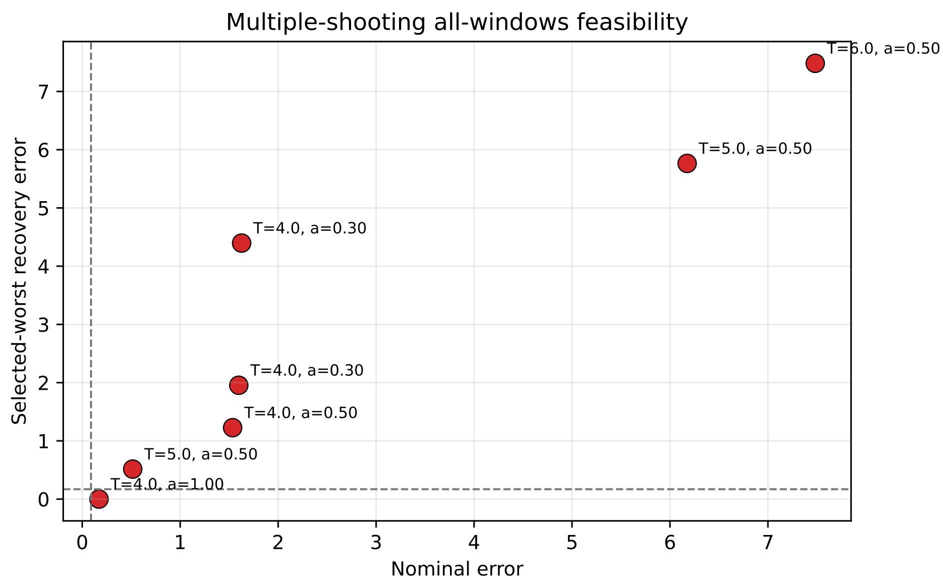

Figure 3: Catalog-derived multiple-shooting feasibility diagnostics. Selected-branch improvement does not remove the nominal and all-mask failure modes.

Table 5: Targeted catalog-derived feasibility attempt with stronger multistart branch recovery. Thresholds were not relaxed.

Transfer time	amax	Segments	Max nfev	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Meets thresholds	Best start	nfev	Runtime (s)
4.0000	0.3000	14	250	0.3055	1.3513	5.4449	False	zero	92	389.4414

Table 6: Selected-outage hard-catalog feasibility-envelope diagnostics. Selected recovery branches are optimized for the listed masks, but all rows fail the nominal threshold and all-mask values are diagnostics, not robustness claims.

Case	Method	Transfer time	amax	Segments	Selected outages	Nominal error	Selected worst error	All-mask diagnostic worst error	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds
selected_branch	multiple_shooting	4.0000	0.7000	14	1	0.3535	0.0360	21.8255	0.7000	0.0000	180	165.8788	False
selected_branch	multiple_shooting	4.0000	1.0000	14	1	0.8020	0.0009	9.3644	1.0000	0.0000	180	259.9800	False
selected_branch	multiple_shooting	4.0000	1.0000	14	3	1.6580	0.0150	8.7872	1.0000	0.0000	180	610.7961	False

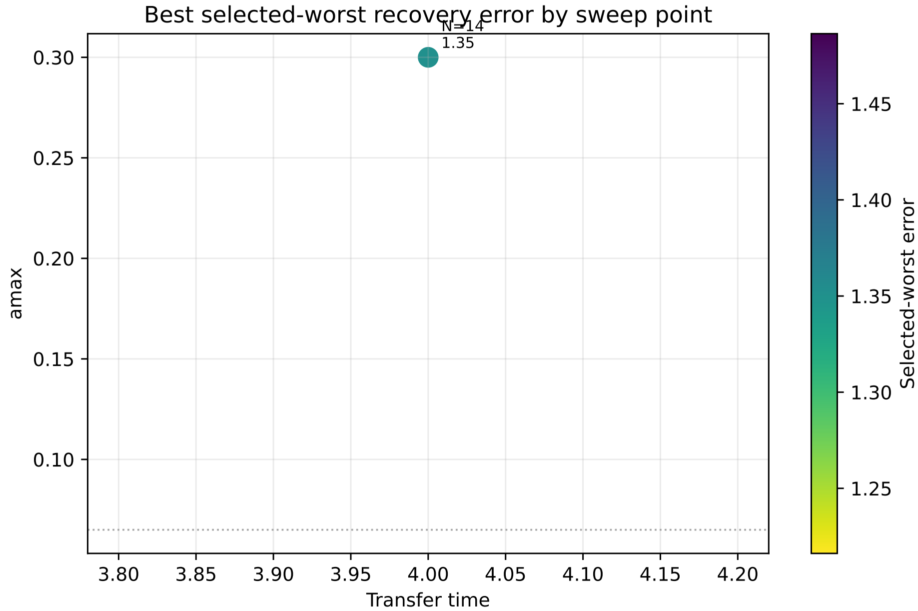

Figure 4: Targeted catalog-derived feasibility attempt. The stronger multistart attempt remains outside the configured thresholds.

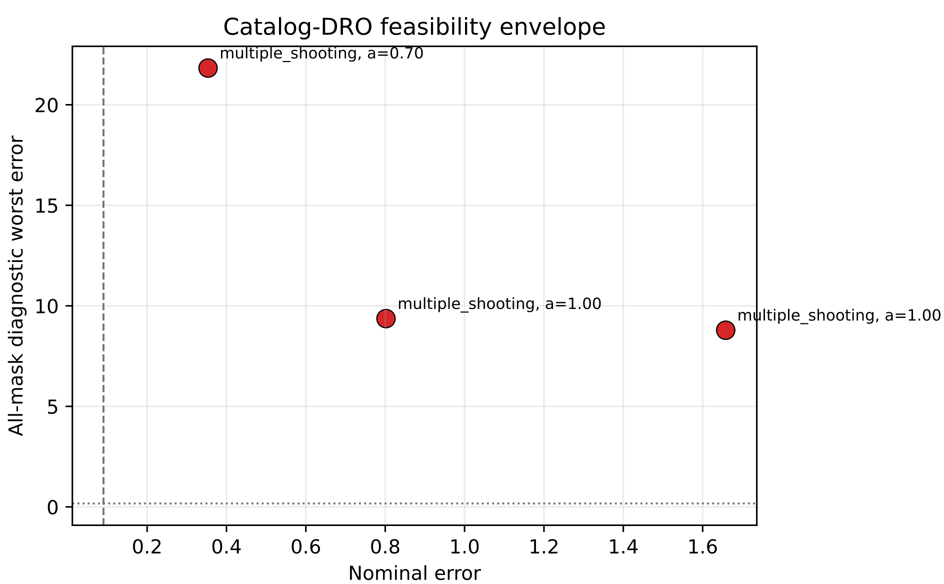

Figure 5: Selected-outage hard-catalog envelope. Small selected-outage errors do not imply robust success because nominal errors fail and all-mask diagnostics remain high.

3 Hard-Catalog Recovery Diagnostics

The locked-nominal, delayed-arrival, and tail-coast packages are continuous-backend diagnostics. They are not QUBO, QAOA, quantum, or discrete-sampler evidence. They are retained here to show the progression from selected-mask overfitting, to locked-nominal branch recovery, to delayed-arrival feasibility, and finally to the scoped fixed-final-time tail-coast row used in the main article.

Table 7: Locked-nominal hard-catalog branch-recovery diagnostics on the NRHO-like to DRO-phase benchmark. The nominal control is solved once and frozen; each selected missed-thrust branch is optimized independently after the outage. The all-mask column is a diagnostic over every configured mask, not a robustness claim. The `locked_hard_single_min6_selected8` row meets the thresholds but is not optimizer-converged.

Case	Policy	Selected masks	Nominal error	Selected worst error	All-mask diagnostic worst	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds
locked_hard_all_single	all_single	14	0.0131	0.8044	8.5410	1.0000	0.0000	955	575.7630	False
locked_hard_nominal_only	0	0	0.0131	0.0131	8.5410	1.0000	0.0000	71	40.6265	True
locked_hard_selected1	1	1	0.0131	0.1973	8.1384	1.0000	0.0000	161	96.3449	False
locked_hard_selected3	3	3	0.0131	0.1973	6.2972	1.0000	0.0000	401	247.4372	False
locked_hard_single_min6_selected8	8	8	0.0131	0.1580	1.4982	1.0000	0.0000	823	495.8345	True

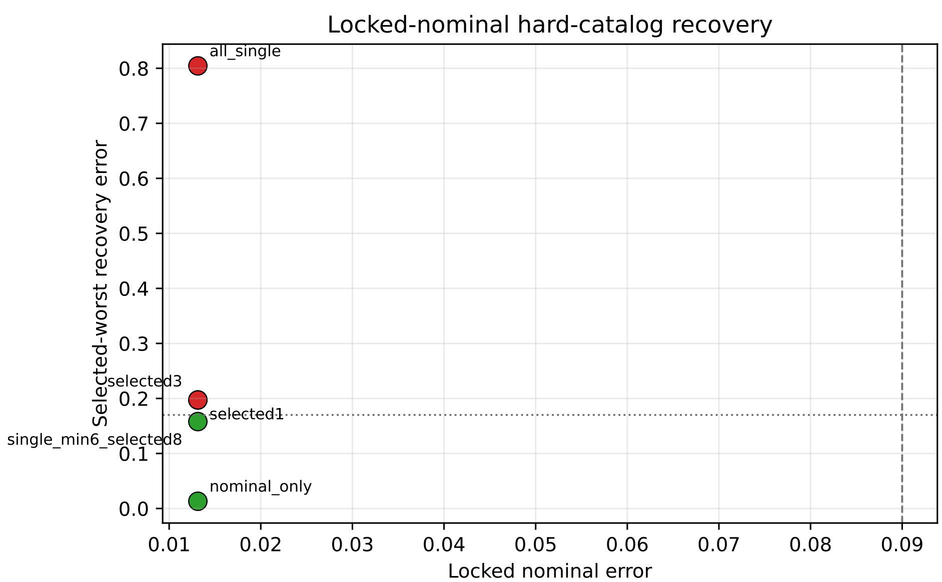

Figure 6: Locked-nominal hard-catalog branch recovery. With the nominal control held feasible, a bounded one-segment, six-recovery-segment selected subset meets the thresholds, while selected one/two-segment and all-single-outage scopes remain above the robust threshold.

Table 8: Delayed-arrival locked-nominal hard-catalog branch-recovery diagnostics. Branch duration is $(N + h)\Delta t$ and the branch target is the original target propagated ballistically by $h\Delta t$. The h6 portfolio row optimizes all 14 one-segment masks, charges all residual variants, and is delayed-arrival feasibility evidence only.

Case	Policy	Selected masks	Horizon segments	Portfolio	Variants	Nominal original error	Nominal delayed coast error	Selected delayed worst error	All-mask delayed diagnostic worst	Terminal weight	Control weight	Smooth weight	Continuity weight	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds	Nominal optimizer success	Branch optimizer ran	Branch optimizer all success
delayed_hard_all_single_h4_regularized	all_single	14	4	False	["configured"]	0.0131	0.0255	1.3990	1.3990	4.0000	0.0100	0.0100	0.0000	1.0000	0.0000	1338	1076.4599	False	True	True	False
delayed_hard_all_single_h6_portfolio	all_single	14	6	True	["regularized_001", "terminal_only"]	0.0131	0.0380	0.0040	0.0040	4.0000	0.0100	0.0100	0.0000	1.0000	0.0000	2253	2639.7199	True	True	True	True
delayed_hard_nominal_only_h4	0	0	4	False	["configured"]	0.0131	0.0255	0.0255	7.3102	4.0000	0.0100	0.0100	0.0000	1.0000	0.0000	71	37.4396	True	True	False	False

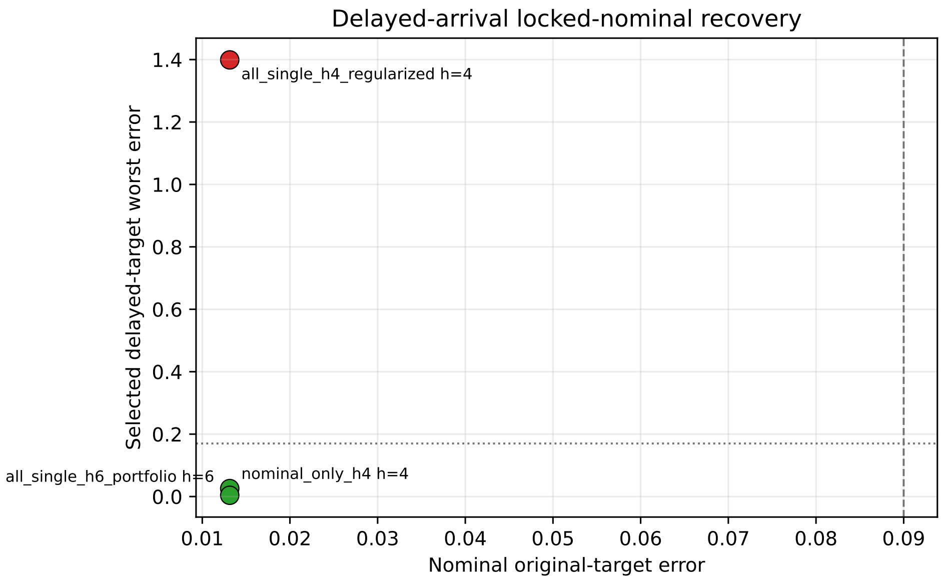

Figure 7: Delayed-arrival locked-nominal hard-catalog recovery. The h4 regularized all-single row fails, while the h6 branch-residual portfolio recovers all one-segment masks against the delayed target.

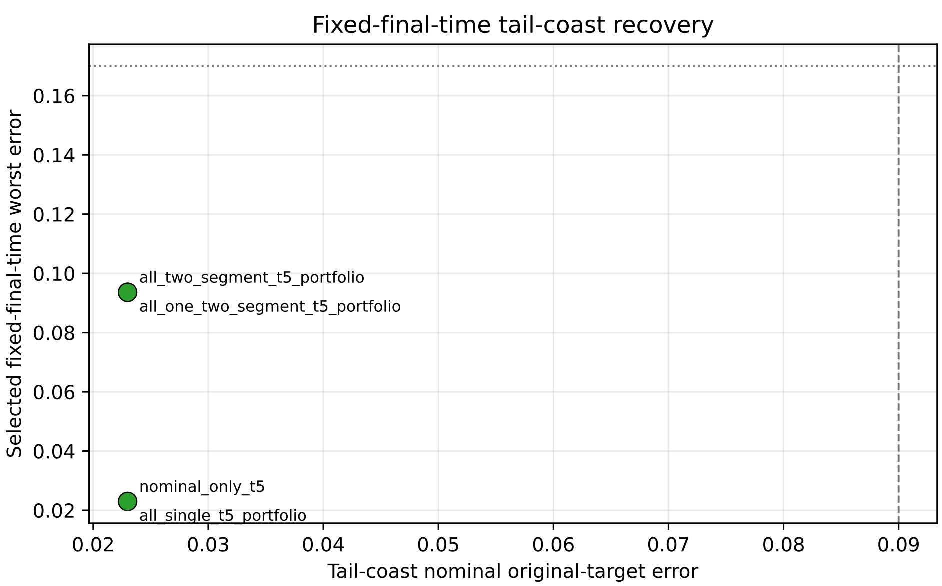

Figure 8: Tail-coast fixed-final-time hard-catalog recovery. The main article reports the table; this visual diagnostic is retained for the recovery-error pattern.

4 Phase-Shift Continuous-Backend Diagnostics

The following tables and figures document the non-teacher phase-shift continuous-backend evidence. These rows show that some normalized CR3BP phase-shift cases are feasible with research-grade continuous backends, but they do not establish certified trajectory-design capability or quantum evidence.

Table 9: Bounded-projected multiple-shooting diagnostics on the non-teacher catalog halo phase-shift target. Controls are projected to $\|u_i\| \leq a_{\max}$ inside the residual and evaluation paths.

Transfer time	amax	Segments	Selected outages	Initialization	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Fuel	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds
0.5000	0.2000	12	1	linear	0.0479	0.0619	0.0752	0.1000	0.2000	0.0000	24	18.0024	True
0.5000	0.2000	8	8	linear	0.0414	0.3378	0.3378	0.1000	0.2000	0.0000	30	73.0455	False
0.5000	0.2000	12	12	linear	0.0396	0.3904	0.3904	0.1000	0.2000	0.0000	20	141.5146	False

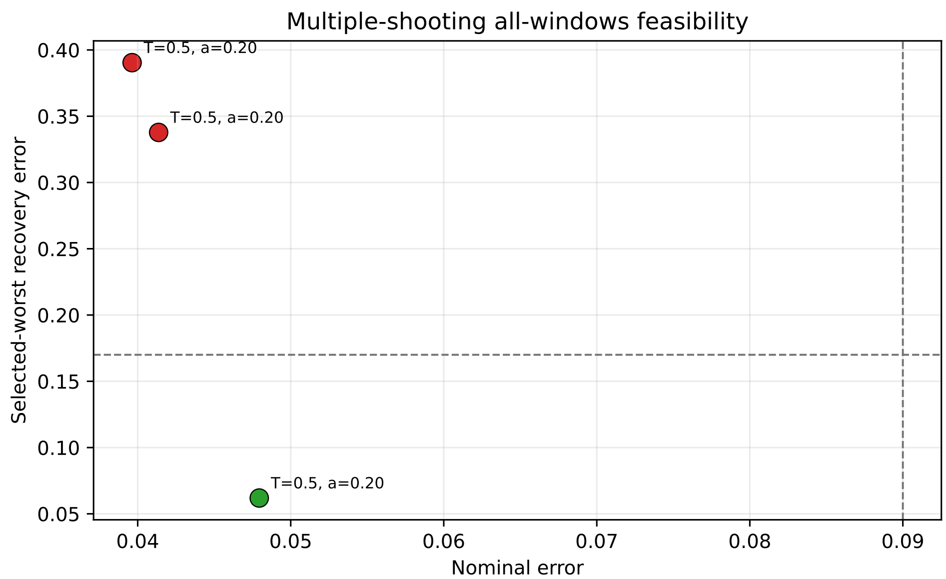

Figure 9: Bounded-projected multiple-shooting diagnostics for the non-teacher phase-shift target.

Table 10: Bounded-projected multiple-shooting phase-suite diagnostics on non-teacher catalog halo phase-shift targets. The selected-branch rows optimize one outage branch; the all-outage-selected row optimizes all one-segment outage branches and remains infeasible.

Case	Phase time	Transfer time	amax	Segments	Selected outages	Nominal error	Selected worst error	All-mask diagnostic worst error	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds
all_outage_selected_diagnostic	0.3000	0.5000	0.2000	12	12	0.0396	0.3581	0.3581	0.2000	0.0000	40	366.6997	False
selected_branch	0.2000	0.5000	0.2000	12	1	0.1939	0.2168	0.2288	0.2000	0.0000	22	12.6848	False
selected_branch	0.3000	0.5000	0.2000	12	1	0.0479	0.0619	0.0752	0.2000	0.0000	24	17.5261	True
selected_branch	0.4000	0.5000	0.2000	12	1	0.0260	0.0083	0.0274	0.2000	0.0000	40	40.3375	True

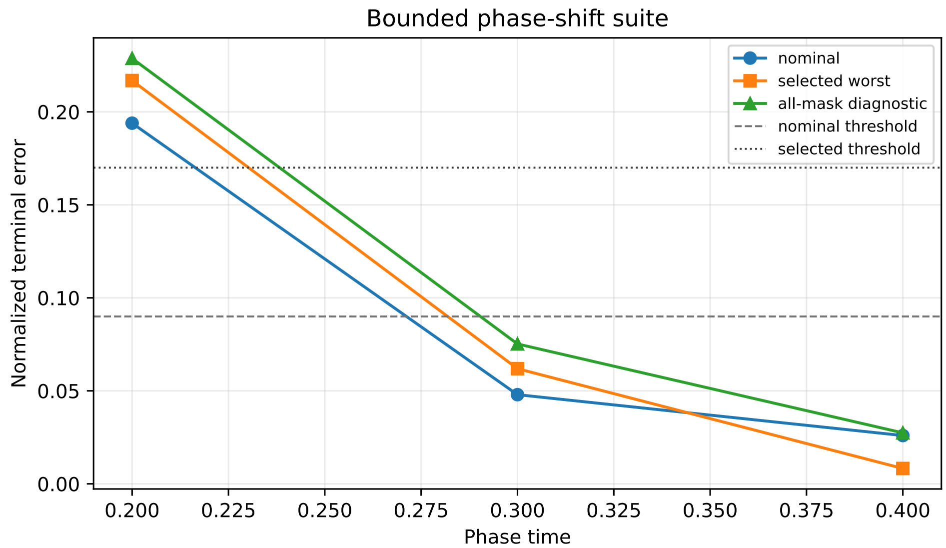

Figure 10: Bounded phase-suite diagnostics on non-teacher catalog halo phase-shift targets.

Table 11: Robust-margin suite on non-teacher catalog halo phase-shift targets. The thrust-margin rows optimize one selected one-segment outage branch and report all one-segment masks diagnostically. The all-single-outage rows optimize every one-segment outage mask for that row; this does not include two-segment outage families.

Group	Phase time	Transfer time	amax	Segments	Max nfev	Selected outages	Outage masks	All selected?	Nominal error	Selected worst error	All-mask worst error	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds
All single outages	0.2000	0.5000	0.3000	8	60	8	8	yes	0.0472	0.2523	0.2523	0.3000	0.0000	60	137.8249	False
All single outages	0.3000	0.5000	0.3000	8	60	8	8	yes	0.0555	0.0358	0.0358	0.3000	0.0000	60	128.1735	True
All single outages	0.4000	0.5000	0.3000	8	60	8	8	yes	0.0531	0.0150	0.0150	0.3000	0.0000	57	119.3214	True
Thrust margin	0.2000	0.5000	0.2000	12	80	1	12	no	0.1939	0.2168	0.2288	0.2000	0.0000	22	11.1493	False
Thrust margin	0.2000	0.5000	0.2500	12	80	1	12	no	0.1136	0.1418	0.1543	0.2500	0.0000	27	14.0958	False
Thrust margin	0.2000	0.5000	0.3000	12	80	1	12	no	0.0538	0.0738	0.0863	0.3000	0.0000	30	16.5662	True

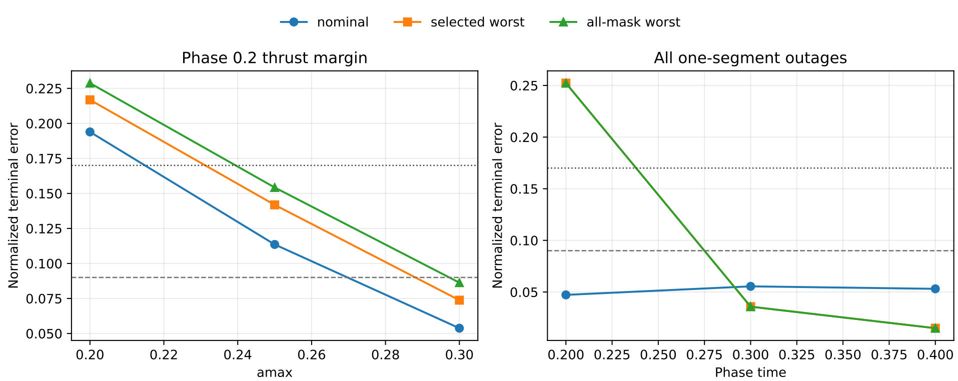

Figure 11: Robust-margin suite summary. The phase-time 0.2 case is thrust-margin sensitive, while higher-thrust smaller cases can meet screening thresholds.

Table 12: Continuation-extension continuous-backend diagnostics on normalized CR3BP catalog halo phase-shift targets. This is a direct multiple-shooting continuation result, not a quantum, QUBO, QAOA, or discrete-sampler result.

Group	Phase time	Warm start	Segments	Outage masks	Max nfev	Nominal error	Selected worst error	All-mask worst error	Optimizer success	MS success	nfev	Runtime (s)	Meets thresholds
all single headroom	0.3000	cold	8	8/8	120	0.0555	0.0351	0.0351	True	True	77	138.7051	True
all single headroom	0.4000	from 0.3	8	8/8	100	0.0531	0.0139	0.0139	True	True	61	120.8934	True
two segment budget test	0.3000	cold	6	11/11	120	0.0592	0.0690	0.0690	True	True	74	106.8293	True
two segment budget test	0.2000	from 0.3	6	11/11	160	0.0651	0.2193	0.2193	False	False	160	253.8218	False
two segment budget test	0.3000	cold	8	15/15	120	0.0612	0.0435	0.0435	True	True	76	243.8341	True

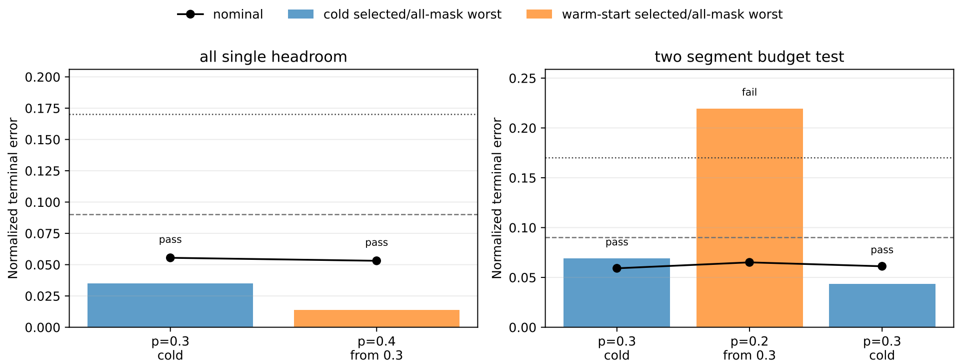

Figure 12: Continuation-extension continuous-backend diagnostics. Warm-start sidecars record nominal-control provenance.

Table 13: Compact bounded projected trapezoidal direct-collocation baseline on the non-teacher catalog halo phase-shift suite. Rows optimize one selected outage branch and report all-mask diagnostics separately.

Case	Phase time	Transfer time	amax	Segments	Selected outages	Nominal error	Selected worst error	All-mask diagnostic worst error	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds
selected_branch	0.2000	0.5000	0.2000	12	1	0.4512	0.4794	0.4805	0.2000	0.0000	30	43.4778	False
selected_branch	0.3000	0.5000	0.2000	12	1	0.3714	0.3774	0.3969	0.2000	0.0000	30	48.8181	False
selected_branch	0.4000	0.5000	0.2000	12	1	0.0357	0.0191	0.0602	0.1483	0.0000	11	18.9927	True

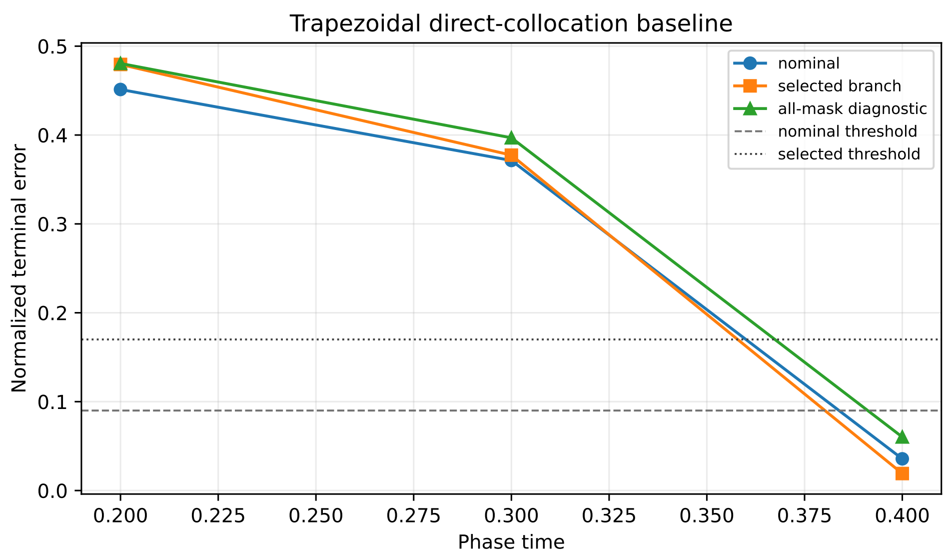

Figure 13: Direct-collocation baseline summary. The compact transcription solves the easiest phase-suite row and fails the harder rows.

Table 14: Constant-control Hermite–Simpson continuation baseline/probe on normalized CR3BP catalog halo phase-shift targets. Warm rows reuse persisted nominal-control sidecars. This is continuous-backend evidence only.

Case ID	Group	Phase time	Warm start	Segments	Outage masks	Max nfev	Method	Nominal error	Selected worst error	All-mask worst error	Optimizer success	DC success	nfev	Runtime (s)	Meets thresholds
hs_phase_p03_cold	Phase continuation (single outage)	0.3000	cold	8	3/8	50	hermite_simpson	0.0376	0.0287	0.0901	False	True	50	145.1586	True
hs_phase_p02_warm_from_p03	Phase continuation (single outage)	0.2000	from 0.3	8	3/8	80	hermite_simpson	0.1069	0.1208	0.1791	False	False	80	249.4395	False
hs_phase_p04_warm_from_p03	Phase continuation (single outage)	0.4000	from 0.3	8	3/8	80	hermite_simpson	0.0179	0.0128	0.0628	True	True	7	25.1659	True
hs_hard_p04_amax02_warm_from_p03	Hard catalog stress probe	0.4000	from 0.3	8	3/8	60	hermite_simpson	0.0183	0.0176	0.0439	True	True	24	74.8323	True

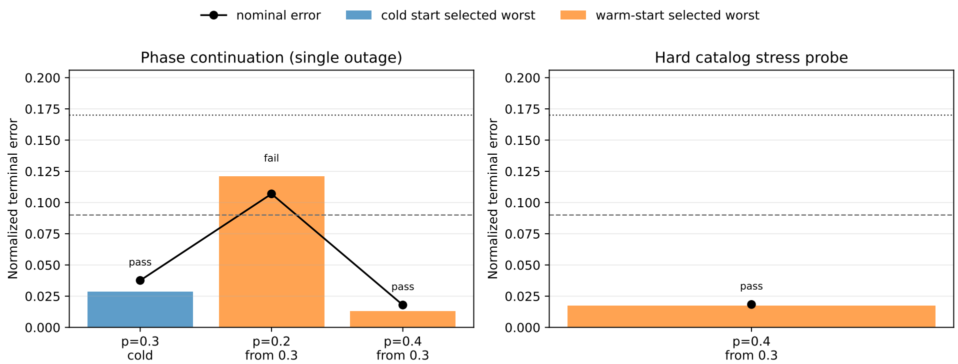

Figure 14: Constant-control Hermite–Simpson continuation baseline/probe summary.

Table 15: Independent-midpoint-control Hermite–Simpson continuation evidence. Warm rows reuse persisted endpoint and midpoint nominal-control sidecars when available. This is continuous-backend evidence only; the catalog-DRO row is a bounded selected-outage negative diagnostic.

Case ID	Group	Phase time	Warm start	Segments	Outage masks	Max nfev	Method	Nominal error	Selected worst error	All-mask worst error	Optimizer success	DC success	nfev	Runtime (s)	Meets thresholds
ihs_phase_p03_cold	Phase continuation (single outage)	0.3000	cold	8	3/8	60	hermite_simpson_midpoint	0.0358	0.0300	0.0849	False	True	60	221.3432	True
ihs_phase_p02_warm_from_p03	Phase continuation (single outage)	0.2000	from 0.3	8	3/8	90	hermite_simpson_midpoint	0.0976	0.0942	0.1694	False	False	90	419.5509	False
ihs_phase_p04_warm_from_p03	Phase continuation (single outage)	0.4000	from 0.3	8	3/8	90	hermite_simpson_midpoint	0.0194	0.0152	0.0605	True	True	7	34.3690	True
ihs_phase_p04_amax02_warm_from_p03	tighter thrust phase shift	0.4000	from 0.3	8	3/8	80	hermite_simpson_midpoint	0.0197	0.0188	0.0532	True	True	17	73.7907	True
ihs_catalog_dro_tf4_amax1_selected1	hard catalog selected outage	0.0000	cold	8	1/15	50	hermite_simpson_midpoint	4.2644	11.0843	11.0843	False	False	50	50.2186	False

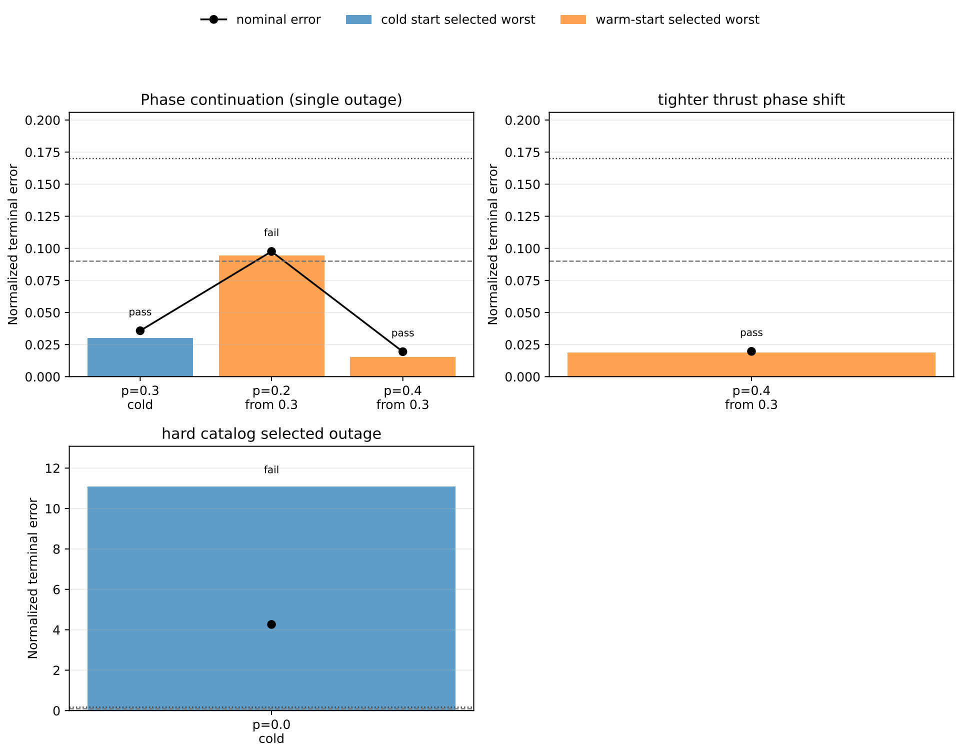

Figure 15: Independent-midpoint-control Hermite–Simpson continuation summary. The package strengthens the continuous-backend comparison but keeps the phase-time 0.2 and catalog-DRO diagnostics unresolved.

5 Phase-Shift Schedule Diagnostics and Small-Seed Evidence

The unrepaired $N = 8$ phase-shift benchmark, deterministic dense-repair ablation, and legacy 10-seed duty-cycle-prior package explain the schedule-encoding failure mode behind the main 30-seed result. The unrepaired samplers can choose sparse late-thrust schedules that do not refine, while dense repair collapses methods to the same all-windows envelope. The 10-seed cardinality-prior package is retained as historical small-seed evidence; the main article uses the 30-seed package for claims.

Table 16: Catalog halo phase-shift benchmark. The target is generated by ballistic CR3BP phase propagation of the JPL source state, not by a low-thrust teacher.

Benchmark	Method	Comparison group	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
catalog_halo_phase_shift	random	method_comparison	0.2663	0.3042	0.3042	0.3750	52.0000	0.0000	52.0000	0.0000
catalog_halo_phase_shift	cross_entropy	method_comparison	0.2663	0.3042	0.3042	0.3750	56.0000	0.0000	56.0000	0.0000
catalog_halo_phase_shift	genetic	method_comparison	0.2663	0.3042	0.3042	0.3750	52.0000	0.0000	52.0000	0.0000
catalog_halo_phase_shift	true_sa	method_comparison	0.2663	0.3042	0.3042	0.3750	52.0000	0.0000	52.0000	0.0000
catalog_halo_phase_shift	surrogate_qubo_sa	method_comparison	0.2663	0.3042	0.3042	0.3750	20.0000	32.0000	52.0000	0.0000
catalog_halo_phase_shift	qaoa_statevector	method_comparison	0.2242	0.2625	0.2650	0.5000	20.0000	32.0000	52.0000	0.0000
catalog_halo_phase_shift	all_windows_continuous	method_comparison	0.0418	0.0750	0.0887	1.0000	30.0000	0.0000	30.0000	1.0000

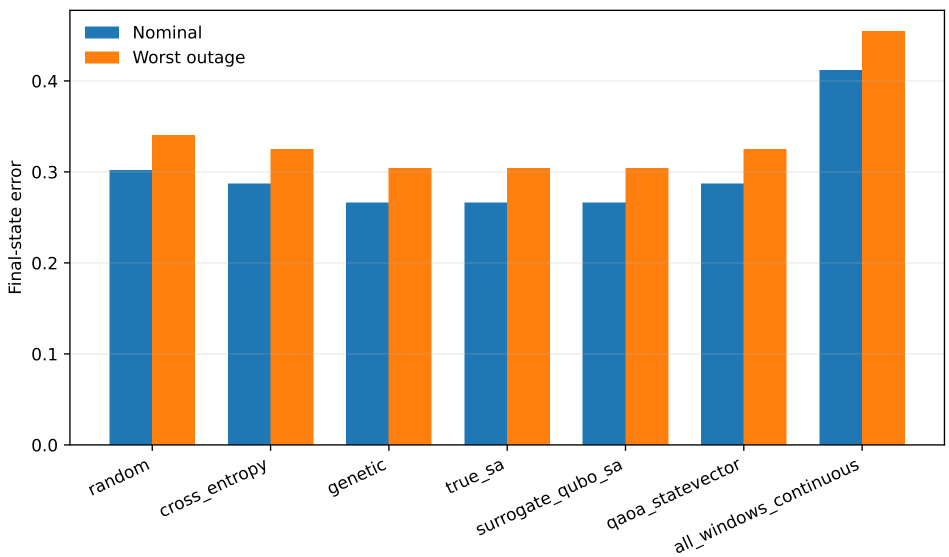

Figure 16: Unrepaired phase-shift method comparison. The all-windows continuous baseline succeeds where sparse binary schedules do not.

Table 17: Catalog halo phase-shift benchmark with deterministic dense-schedule repair. The table exposes the solver schedule, refinement candidate schedule, repaired refined schedule, and active fractions to show that repaired methods collapse to the same all-windows refinement envelope.

Benchmark	Method	Schedule repair	Refined schedule source	Comparison group	Solver best schedule	Refine candidate schedule	Refined schedule
catalog_halo_phase_shift	random	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111
catalog_halo_phase_shift	cross_entropy	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111
catalog_halo_phase_shift	genetic	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	00000111	00000111	11111111
catalog_halo_phase_shift	true_sa	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111
catalog_halo_phase_shift	surrogate_qubo_sa	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	00000111	00000111	11111111
catalog_halo_phase_shift	qaoa_statevector	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111
catalog_halo_phase_shift	all_windows_continuous	identity	original solver candidate schedule	method_comparison	11111111	11111111	11111111

Method	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Solver best active fraction	Refine candidate active fraction	Refined active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
random	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	52.0000	0.0000	52.0000	1.0000
cross_entropy	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	56.0000	0.0000	56.0000	1.0000
genetic	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	52.0000	0.0000	52.0000	1.0000
true_sa	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	52.0000	0.0000	52.0000	1.0000
surrogate_qubo_sa	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	20.0000	32.0000	52.0000	1.0000
qaoa_statevector	0.0418	0.0750	0.0887	0.5000	0.5000	1.0000	20.0000	32.0000	52.0000	1.0000
all_windows_continuous	0.0418	0.0750	0.0887	1.0000	1.0000	1.0000	30.0000	0.0000	30.0000	1.0000

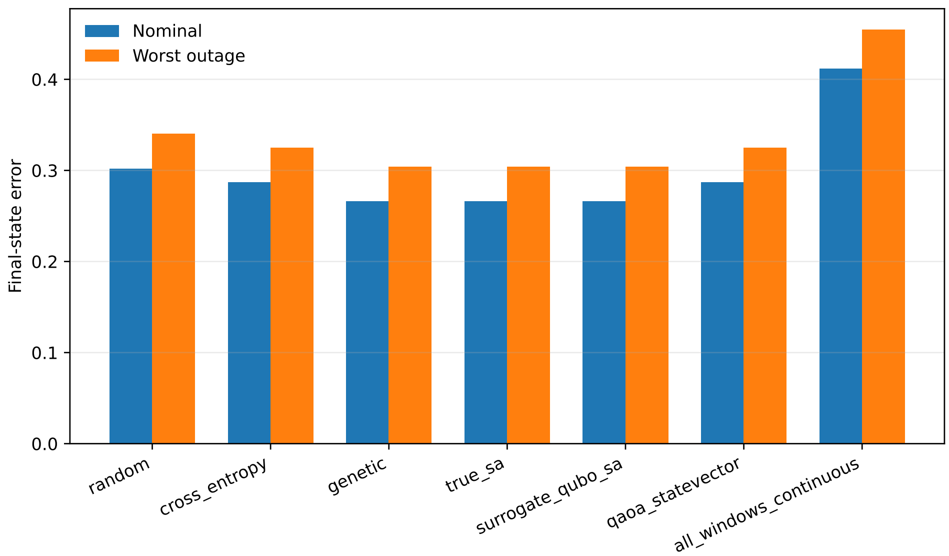

Figure 17: Dense-repair phase-shift comparison. The repair is an encoding diagnostic, not a competitive method-ranking result.

Table 18: Legacy 10-seed $N = 12$ duty-cycle-prior phase-shift benchmark. QUBO/QAOA totals include 96 shared QUBO-training evaluations and 24 post-QUBO true candidate evaluations.

Benchmark	Method	Schedule repair	Refined schedule source	Comparison group	Solver best schedule	Refine candidate schedule	Refined schedule	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Solver best active fraction	Active target penalty	Refine candidate active fraction	Refined active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
catalog_halo_phase_shift	random	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0954	0.1214	0.1313	0.8333	0.0833	0.8333	0.8333	120.0000	0.0000	120.0000	0.2000
catalog_halo_phase_shift	cross_entropy	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	genetic	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0675	0.0926	0.1020	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	true_sa	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	surrogate_qubo_sa	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	24.0000	96.0000	120.0000	1.0000
catalog_halo_phase_shift	qaoa_statevector	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0942	0.1201	0.1292	0.8333	0.0833	0.8333	0.8333	24.0000	96.0000	120.0000	0.4000
catalog_halo_phase_shift	all_windows_continuous	identity	original solver candidate schedule	method_comparison	111111111111	111111111111	111111111111	0.0386	0.0601	0.0690	1.0000	0.0833	1.0000	1.0000	19.0000	0.0000	19.0000	1.0000

Table 19: Legacy 10-seed duty-cycle-prior recovery fuel and refinement effort. Fuel is reported in normalized acceleration-time units.

Method	Schedule repair	Refined schedule	Refined schedule source	Nominal fuel	Mean recovery fuel	Max recovery fuel	Refinement nfev	Recovery success
random	identity	mixed	original solver candidate schedule	0.0833	0.0750	0.0750	26.0000	0.2000
cross_entropy	identity	mixed	original solver candidate schedule	0.0917	0.0833	0.0833	20.0000	1.0000
genetic	identity	mixed	original solver candidate schedule	0.0917	0.0833	0.0833	35.0000	1.0000
true_sa	identity	mixed	original solver candidate schedule	0.0917	0.0833	0.0833	20.0000	1.0000
surrogate_qubo_sa	identity	mixed	original solver candidate schedule	0.0917	0.0833	0.0833	20.0000	1.0000
qaoa_statevector	identity	mixed	original solver candidate schedule	0.0833	0.0750	0.0750	27.0000	0.4000
all_windows_continuous	identity	111111111111	original solver candidate schedule	0.1000	0.0917	0.0917	19.0000	1.0000

Table 20: Legacy 10-seed duty-cycle-prior QUBO surrogate diagnostics.

seed	ridge	train_rmse	val_rmse	train_r2	val_r2	val_spearman	val_pairwise_order_accuracy	val_top_k	val_top_k_recall	n_train	n_val	shared_qubo_training_evaluations	qubo_training_true_evaluations	equal_total_budget
11	0.0001	0.0001	0.1056	1.0000	0.9961	0.9896	0.9674	5	1.0000	72	24	96	96	True
23	0.0001	0.0001	0.5663	1.0000	0.9220	0.9000	0.8804	5	0.6000	72	24	96	96	True
37	0.0001	0.0001	0.1579	1.0000	0.9923	0.9878	0.9638	5	1.0000	72	24	96	96	True
41	0.0001	0.0002	0.5265	1.0000	0.9405	0.9470	0.9130	5	0.8000	72	24	96	96	True
53	0.0001	0.0001	0.2576	1.0000	0.9601	0.9487	0.9275	5	1.0000	72	24	96	96	True
67	0.0001	0.0001	0.4435	1.0000	0.9677	0.9652	0.9348	5	0.6000	72	24	96	96	True
79	0.0001	0.0001	0.2407	1.0000	0.9848	0.9756	0.9491	5	0.8000	72	24	96	96	True
83	0.0001	0.0001	0.2601	1.0000	0.9744	0.9557	0.9275	5	0.8000	72	24	96	96	True
97	0.0001	0.0001	0.1768	1.0000	0.9916	0.9791	0.9529	5	1.0000	72	24	96	96	True
101	0.0001	0.0001	0.2653	1.0000	0.9591	0.9748	0.9384	5	1.0000	72	24	96	96	True

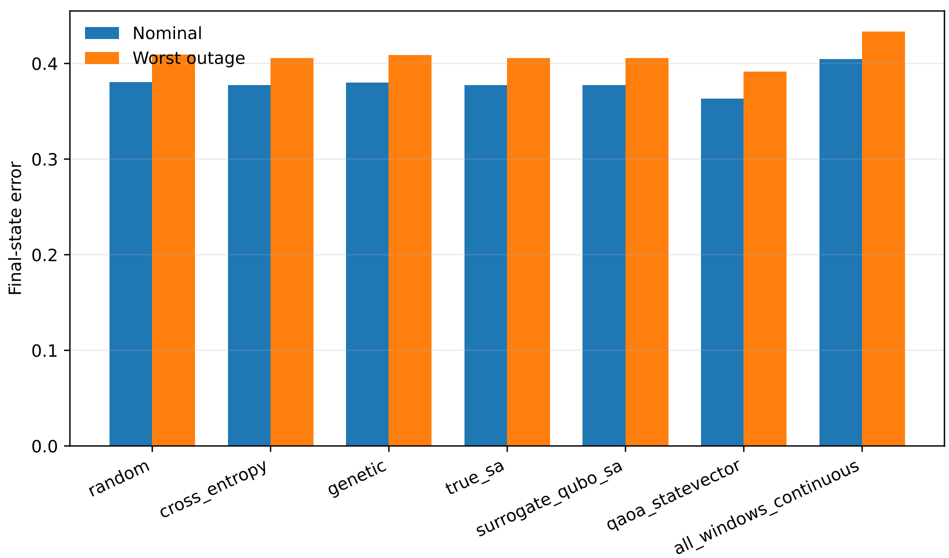

Figure 18: Legacy 10-seed duty-cycle-prior method comparison. The main article uses the 30-seed package for statistical claims.

6 Cardinality and Fuel Ablations

The duty-cycle-prior result is positive but narrow. The high-duty ablations show that one-coast schedules belong to a broad feasible region, and the fuel sweep shows that stronger all-windows continuous fuel regularization can reduce fuel while retaining feasibility. These checks prevent overclaiming one-coast schedules as Pareto-superior.

Table 21: Exhaustive high-duty cardinality ablation for the $N = 12$ phase-shift benchmark.

k_active	schedules_refined	success_count	all_mask_success_count	best_nominal_error	median_nominal_error	best_selected_worst_error	median_selected_worst_error	best_all_mask_worst_error	median_all_mask_worst_error	best_nominal_fuel	median_nominal_fuel	best_schedule_by_selected_worst	min_fuel_feasible_schedule
10	66	41	41	0.0798	0.0874	0.1056	0.1132	0.1146	0.1230	0.0833	0.0833	111111111100	111111011110
11	12	12	12	0.0588	0.0617	0.0833	0.0864	0.0923	0.0956	0.0917	0.0917	111111111110	111111111011
12	1	1	1	0.0386	0.0386	0.0601	0.0601	0.0690	0.0690	0.1000	0.1000	111111111111	111111111111

Table 22: One-coast method selections compared with the refined one-coast frontier.

method	runs	one_coast_runs	one_coast_success_runs	one_coast_frontier_runs	near_best_selected_runs	median_method_selected_worst_error	median_method_nominal_fuel	median_exhaustive_selected_worst_error	median_exhaustive_nominal_fuel	median_selected_gap_to_best_one_coast	dominant_one_coast_schedules
surrogate_qubo_sa	10	10	10	0	0	0.0941	0.0917	0.0941	0.0917	0.0108	011111111111:7; 101111111111:3
cross_entropy	10	10	10	0	1	0.0941	0.0917	0.0941	0.0917	0.0108	011111111111:6; 110111111111:2; 111110111111:1; 111011111111:1
genetic	10	10	10	0	1	0.0926	0.0917	0.0926	0.0917	0.0093	101111111111:5; 110111111111:3; 011111111111:1; 111101111111:1
true_sa	10	10	10	0	0	0.0941	0.0917	0.0941	0.0917	0.0108	011111111111:6; 101111111111:4
qaoa_statevector	10	4	4	1	2	0.0887	0.0917	0.0887	0.0917	0.0054	111011111111:2; 111101111111:1; 111111111110:1
random	10	1	1	0	0	0.0926	0.0917	0.0926	0.0917	0.0093	101111111111:1

Table 23: One-coast refined schedule diagnostics for the duty-cycle-prior ablation.

schedule	coast_windows_zero_based	refinement_success	one_coast_pareto_frontier	refined_nominal_error	refined_selected_worst_error	refined_all_mask_worst_error	refined_nominal_fuel	refinement_nfev
011111111111	0	True	False	0.0690	0.0941	0.1018	0.0917	20
101111111111	1	True	False	0.0675	0.0926	0.1022	0.0917	35
110111111111	2	True	False	0.0670	0.0922	0.1015	0.0917	35
111011111111	3	True	False	0.0643	0.0892	0.0986	0.0917	35
111101111111	4	True	False	0.0633	0.0881	0.0975	0.0917	35
111110111111	5	True	False	0.0628	0.0876	0.0967	0.0917	35
111111011111	6	True	False	0.0601	0.0837	0.0941	0.0917	16
111111101111	7	True	False	0.0606	0.0852	0.0944	0.0917	35
111111110111	8	True	False	0.0601	0.0847	0.0938	0.0917	35
111111111011	9	True	True	0.0603	0.0848	0.0939	0.0917	35
111111111101	10	True	True	0.0596	0.0841	0.0931	0.0917	35
111111111110	11	True	True	0.0588	0.0833	0.0923	0.0917	35

Table 24: All-windows fuel-regularization sweep for the $N = 12$ phase-shift benchmark.

fuel_residual_weight	refinement_success	all_mask_success	refined_nominal_error	refined_selected_worst_error	refined_all_mask_worst_error	refined_nominal_fuel	refinement_nfev
0.0180	True	True	0.0386	0.0601	0.0690	0.1000	19
0.0300	True	True	0.0468	0.0711	0.0791	0.0962	18
0.0500	True	True	0.0773	0.1100	0.1122	0.0846	15
0.0800	False	False	0.1204	0.1491	0.1564	0.0700	10
0.1200	False	False	0.1891	0.2128	0.2183	0.0504	7
0.2000	False	False	0.2755	0.2892	0.2915	0.0268	7

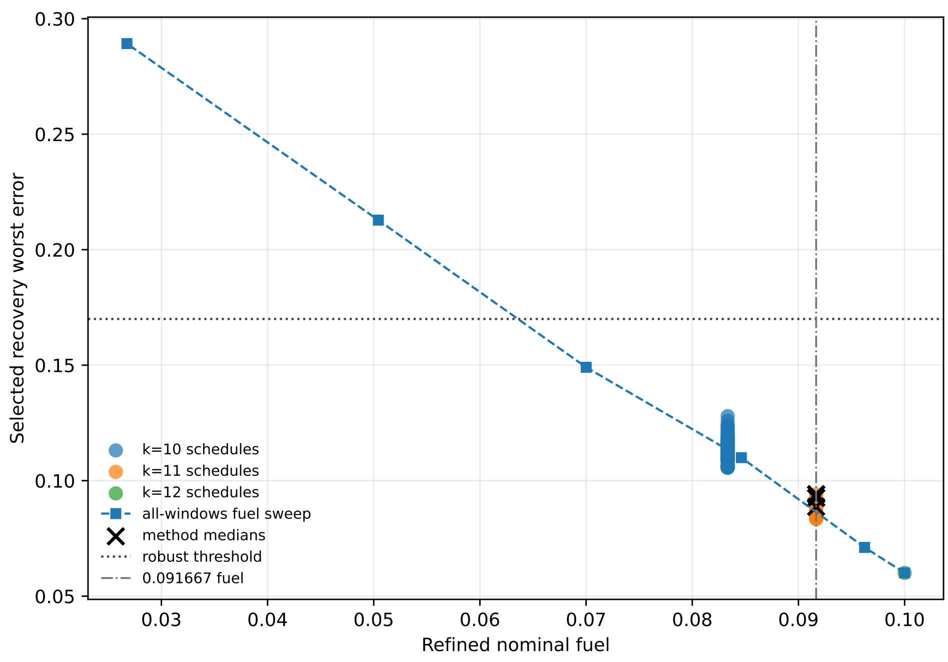

Figure 19: Fuel-error Pareto view for the duty-cycle-prior ablation. Lower fuel is achievable with all-windows fuel regularization while retaining feasibility, so the one-coast result should be read as a schedule-structure feasibility result rather than a fuel-optimality result.

7 Focused QAOA/QUBO Raw Summary and Legacy QAOA Artifacts

The main article reports the 30-seed QAOA/QUBO statistical diagnostics. The raw 30-seed method summary is retained here. Legacy 10-seed QAOA-depth artifacts remain in the repository for provenance but are not the current statistical evidence.

Table 25: QAOA depth and angle-optimization ablation on the $N = 12$ duty-cycle-prior phase-shift benchmark. All rows use 96 shared QUBO-training trajectory evaluations and 24 true post-QUBO candidate evaluations per seed.

method	runs	refinement_success_rate	refined_nominal_error_median	refined_selected_worst_error_median	refined_nominal_fuel_median	solver_true_evaluations_median	shared_qubo_training_evaluations_median	runtime_seconds_median
surrogate_qubo_sa	30	0.9333	0.0690	0.0941	0.0917	24.0000	96.0000	0.2952
exact_qubo_top24	30	0.9333	0.0690	0.0941	0.0917	24.0000	96.0000	0.2967
qaoa_random_p1	30	0.5000	0.0800	0.1055	0.0875	24.0000	96.0000	0.3222
qaoa_optimized_p1	30	0.7000	0.0682	0.0933	0.0917	24.0000	96.0000	0.3212
qaoa_optimized_p2	30	0.9667	0.0690	0.0941	0.0917	24.0000	96.0000	0.3491

Table 26: Legacy 10-seed QAOA-depth ablation retained for provenance. The manuscript's QAOA-depth statistical result uses the 30-seed package.

method	runs	refinement_success_rate	refined_nominal_error_median	refined_selected_worst_error_median	refined_nominal_fuel_median	solver_true_evaluations_median	shared_qubo_training_evaluations_median	runtime_seconds_median
surrogate_qubo_sa	10	1.0000	0.0682	0.0933	0.0917	24.0000	96.0000	0.3912
exact_qubo_top24	10	0.9000	0.0690	0.0941	0.0917	24.0000	96.0000	0.3957
qaoa_random_p1	10	0.3000	0.0948	0.1203	0.0833	24.0000	96.0000	0.4140
qaoa_optimized_p1	10	0.6000	0.0690	0.0941	0.0917	24.0000	96.0000	0.4098
qaoa_optimized_p2	10	0.9000	0.0690	0.0941	0.0917	24.0000	96.0000	0.4964

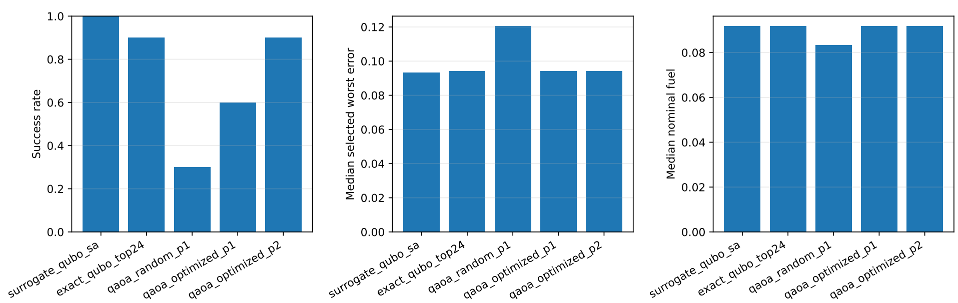

Figure 20: Legacy 10-seed QAOA-depth summary retained as provenance.

8 Teacher-Controlled Diagnostics

The teacher-controlled benchmark is feasible by construction and should not be confused with an operational transfer. It is retained because it diagnoses whether the architecture can recover a known feasible low-thrust target and because the oracle continuous-control diagnostic shows that continuous initialization can dominate the outcome. The teacher-controlled tables are small-seed evidence and do not support a statistically significant method ranking.

Table 27: Teacher-controlled benchmark results. Values are medians across three seeds. QUBO and QAOA totals include the shared QUBO-training trajectory evaluations.

Benchmark	Method	Comparison group	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
teacher_controlled	random	method_comparison	0.0115	0.0137	1.1009	0.3750	52.0000	0.0000	52.0000	0.6667
teacher_controlled	cross_entropy	method_comparison	0.0381	0.2626	0.7695	0.3750	56.0000	0.0000	56.0000	0.3333
teacher_controlled	genetic	method_comparison	0.0213	0.5602	0.8393	0.3125	52.0000	0.0000	52.0000	0.0000
teacher_controlled	true_sa	method_comparison	0.6349	0.9638	16.4017	0.6250	52.0000	0.0000	52.0000	0.0000
teacher_controlled	surrogate_qubo_sa	method_comparison	0.0263	0.2162	0.8145	0.3750	20.0000	32.0000	52.0000	0.3333
teacher_controlled	qaoa_statevector	method_comparison	0.2495	0.2496	1.0213	0.3125	20.0000	32.0000	52.0000	0.3333
teacher_controlled	all_windows_continuous	method_comparison	6.3495	8.4578	17.8387	1.0000	3.0000	0.0000	3.0000	0.0000
teacher_controlled	teacher_seeded_schedule	method_comparison	1.2183	0.9547	1.7652	0.4375	30.0000	0.0000	30.0000	0.0000
teacher_controlled	teacher_controls_oracle_diagnostic	oracle_diagnostic	0.0067	0.0080	1.1910	0.4375	7.0000	0.0000	7.0000	1.0000

Table 28: Teacher-controlled recovery fuel and refinement effort. Fuel is reported in normalized acceleration-time units.

Method	Nominal fuel	Mean recovery fuel	Max recovery fuel	Refinement nfev	Recovery success
random	0.0579	0.0579	0.0579	24.0000	0.6667
cross_entropy	0.0892	0.0813	0.0813	30.0000	0.3333
genetic	0.0679	0.0650	0.0650	30.0000	0.0000
true_sa	0.1592	0.1355	0.1438	16.0000	0.0000
surrogate_qubo_sa	0.0733	0.0650	0.0650	27.0000	0.3333
qaoa_statevector	0.0488	0.0488	0.0488	30.0000	0.3333
all_windows_continuous	0.2595	0.2194	0.2284	3.0000	0.0000
teacher_seeded_schedule	0.1134	0.0793	0.0849	30.0000	0.0000
teacher_controls_oracle_diagnostic	0.0281	0.0272	0.0283	7.0000	1.0000

Table 29: Teacher-controlled QUBO surrogate fit diagnostics.

seed	ridge	train_rmse	val_rmse	train_r2	val_r2	val_spearman	val_pairwise_order_accuracy	val_top_k	val_top_k_recall	n_train	n_val	shared_qubo_training_evaluations	qubo_training_true_evaluations	equal_total_budget
11	0.0001	0.0001	9.4732	1.0000	0.0443	0.7143	0.7857	2	0.5000	24	8	32	32	True
23	0.0001	0.0002	13.1807	1.0000	-1.7931	0.6190	0.7857	2	1.0000	24	8	32	32	True
37	0.0001	0.0001	7.2534	1.0000	0.6466	0.5714	0.6786	2	1.0000	24	8	32	32	True

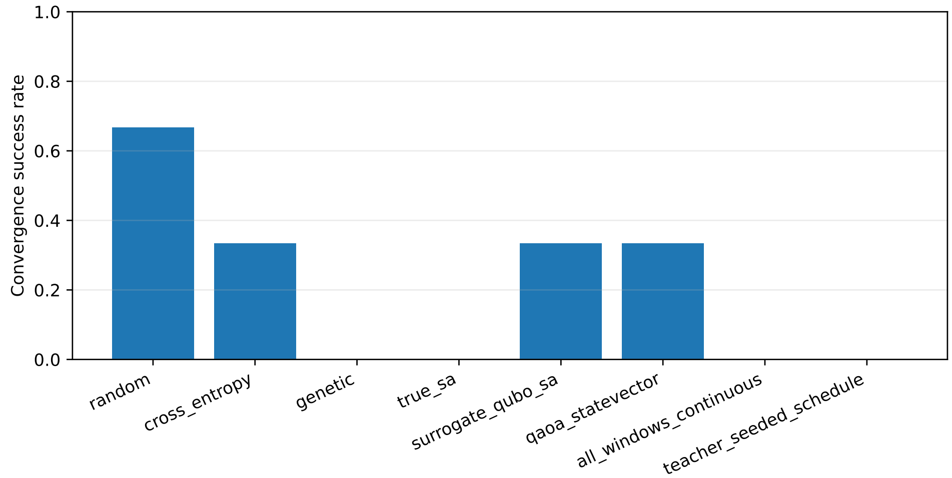

Figure 21: Teacher-controlled selected-branch refinement success rates. The result demonstrates nonzero recovery success but not quantum advantage.

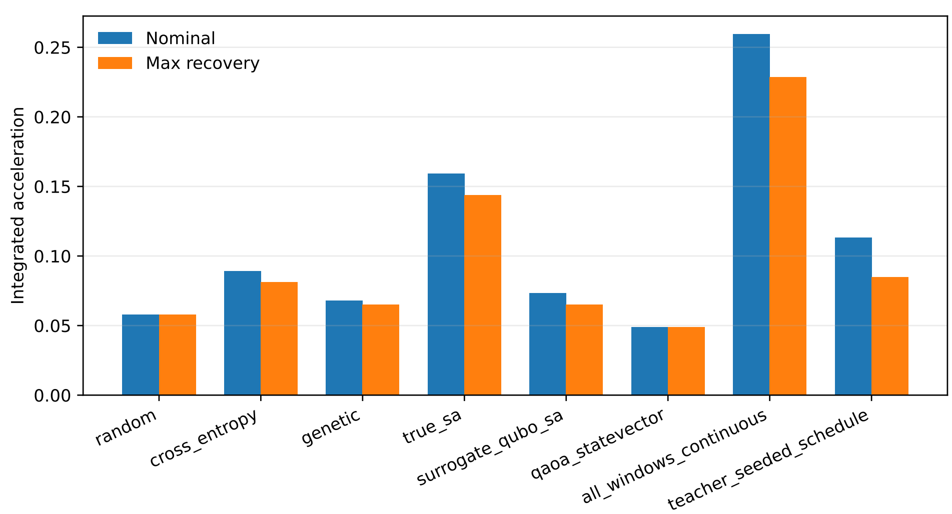

Figure 22: Teacher-controlled nominal and recovery fuel summaries. Fuel differences must be interpreted together with feasibility and missed-thrust error.

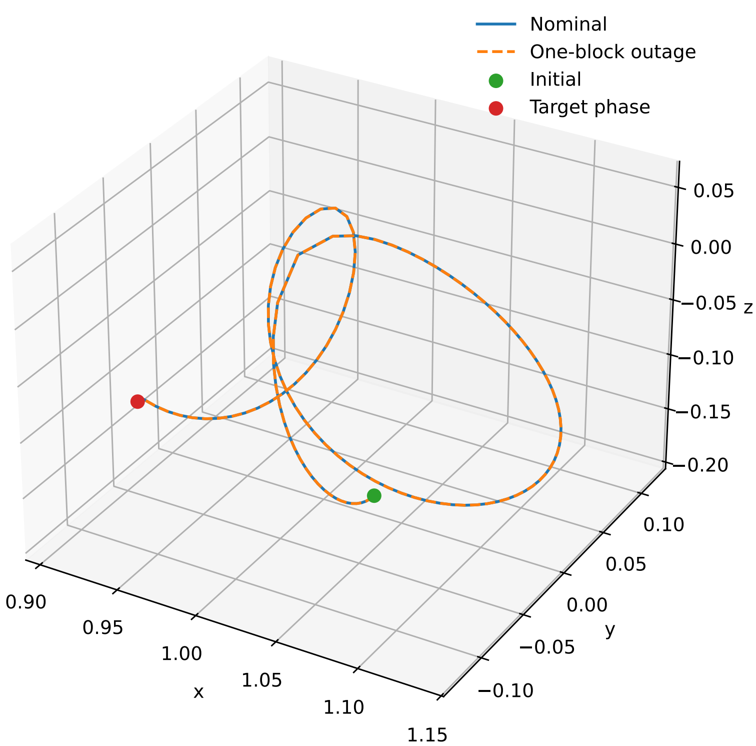

Figure 23: Example trajectory from the teacher-controlled benchmark. The case is a normalized CR3BP algorithm benchmark, not a flight-ready trajectory.

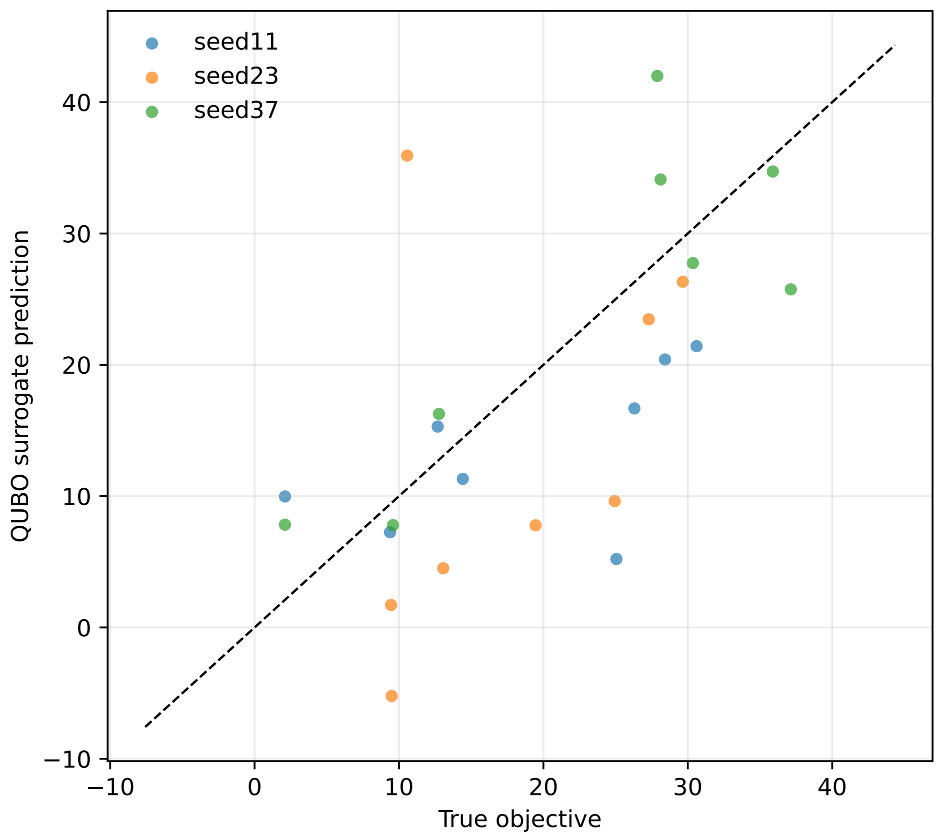

Figure 24: Teacher-controlled QUBO fit diagnostic. Surrogate quality remains mixed across the small teacher-controlled seeds.

9 Artifact and Provenance Notes

The machine-readable manifest `../data/results/artifact_manifest.json` records scoped SHA-256 hashes and byte counts for source, configuration, manuscript, result, table, and figure artifacts. The hashes and byte counts are authoritative for artifact identity. The manifest intentionally excludes itself, virtual environments, caches, logs, and LaTeX auxiliary files. Its `git_head_at_generation` field records the repository state when the manifest was generated; a committed manifest necessarily records the parent/pre-final state rather than the final commit containing the refreshed manifest. Historical experiment metadata may predate the repository or contain dirty-state records, so current submission provenance is verified by clean git status after final commit, manifest checking, tests, and local LaTeX builds.